

1 Quantitative analysis

We use the model to translate the local fiscal multiplier to an aggregate fiscal multiplier. First, we describe our calibration and steady-state results. Then we consider the main quantitative experiment: temporary regional government spending shocks.

Table 1 Benchmark parameters

Parameter	Notation	Value	Target/source
Mean discount factor	$\bar{\beta}$	0.985	Annual real rate = 4%
Dispersion in discount factor	ϵ_{β}	0.005	%Households with $a/y < 1\%$
Labour supply elasticity	$1/\theta$	1.0	Beraja et al. ?
Disutility of labour	ψ	5.8	Hours worked = 42%
Persistence of x	ρ	0.955	Persistence of log-wages = 0.92
Variance of innovation to x	σ_{η}^2	1.5%	Variance of log-wages = 0.04
Per-capita gov. spending	G	0.10	$G/Y = 20\%$
Dividend allocation	ω	0.32	NIPA tables
Elasticity of substitution	ϵ	6	Christiano et al. ?
Price reset probability	λ	0.15	McKay et al. ?
Taylor rule coefficient	ζ	0.0	-
Stock of liquid assets	B^g	1.2 $\times$ annual income	SCFs
Dividend tax	τ_d	0.25	Gourio and Miao ?
Lump-sum transfer	T	0.03	$T/Y = 6\%$
Size of region 1	μ_1	3.1%	Relative county population
Home bias	α	0.58	CFS

1.1 Calibration

Table 7 summarizes our parameter choices. The model period is a quarter. The average discount factor $\bar{\beta}$ is set to match an annual steady-state real interest rate equal to 4%. The utility from leisure parameter ψ is set so that on average households work 42% of their time endowment.¹ Parameter θ that governs the Frisch elasticity of labour supply is set to 1 based on Beraja et al. ?. This value reflects a combined labour supply response at the intensive and extensive margin.

The productivity process is calibrated based on the estimates of Floden and Linde ?. Using our model, we simulate labour income paths and then annualize the simulated data to match a persistence of $\rho = 0.92$ and $\sigma_{\eta}^2 = 0.04$. We set steady-state government spending G to match a government spending to income ratio equal to 20%. The lump-sum transfer T is equal to 6% of annual income (?). We set the elasticity of substitution $\epsilon = 6$ based on Christiano et al. ?. The probability of changing price $\lambda = 0.15$ is based on McKay et al. ?.²

¹In the panel study of income dynamics, prime-age, full-time employed males work around 2,200 h per year. We normalize this value by a time endowment of 5,200 h per year.

²This is on the lower end of the empirical estimates, but it is supported by the absence of inflation pressures during and in the aftermath of the Recovery Act episode.

As Kaplan and Violante ? have shown, households use primarily liquid assets to adjust their consumption. Moreover, Carroll et al. ? have shown that a model that matches the degree of inequality in liquid financial assets generates marginal propensities to consume closer to the empirical estimates. Hence, we calibrate the debt-to-income ratio to match the empirical ratio of liquid assets to income and the discount factor dispersion, represented by the parameter ϵ_β , to match the fraction of households whose liquid assets are less than 1% of their annual income. We use data from the survey of consumer finances (SCFs) for period 1998-2007. We define liquid assets following Kaplan and Violante ?. In particular, liquid financial assets are cash, checking accounts, savings accounts, money market accounts, and stocks net of credit card debt. There are no data in the SCF on households' cash holdings. As a result, we increase liquid asset holdings by a factor of 1.04 (see the appendix in Kaplan and Violante, 2014). In 2009 prices, the average (median) household owns \$94,443(\$3,149) in liquid assets. Average (median) household income is \$78,500(\$46,564). As a result, we target an annual debt-to-income ratio of 1.20 . We also find that the fraction of households whose liquid assets are less than 1% of their annual income is 29%.

Based on the national income accounts, we find that for the period 2007-08, the undistributed corporate profits were, on average, \$394 billion, while dividend income was \$809 billion. Thus, undistributed corporate profits as a share of total dividends (both undistributed and distributed as income) is around 32%. Therefore, we use a value of $\omega = 0.32$, which is slightly lower than the value of 0.33 used by Kaplan et al. ?. The dividend tax equals 0.25 based on Gourio and Miao ?.

There are two regions in the model: one small region and the rest of the economy. Consistent with our empirical exercise, we interpret region 1 as a county. According to our analysis (presented in Supplementary Material, Appendix E), regional spillovers stabilize for geographical regions larger than 1,000 miles, which roughly corresponds to the size of a U.S. Census region.³ Thus, we set $\mu_1 = 0.031$ which represents the average size of a county relative to the Census region (weighted by county-level income in 2010).

The second regional parameter to calibrate is the preference of the final good firm for home versus foreign inputs, α . We calibrate home bias using direct evidence on shipments of goods from the CFS for 2012. As mentioned, the data include information on commodities shipped, their value, weight, and the origin and destination of the shipments across U.S. states. For every state, we compute the total value of shipments that originated and shipped inside the state, as well as the total value of shipments that originated in the state but were exported to other states. We find that on average 42% of shipments of goods stay within state borders.

Next, we note that a large fraction of spending is on services. According to the 2010 National Income and Product Accounts (NIPA) tables, services absorb 68% of total spending. Services are traditionally considered non-tradable, but a substantial fraction can be delivered over a distance (e.g. financial or legal services). Gervais and Jensen ?

³The U.S. Census Bureau considers four regions: the Northeast, the Midwest, the South, and the West.

document that the U.S. was exporting around 30% of its services in 2007. To account for both goods and services, we estimate home bias as $(1 - 0.68) \times 0.42 + 0.68 \times 0.70 = 0.61$.

To calibrate a county-level home bias, we use the state-level home bias and the relative populations of these two geographical areas. Let $\mathcal{M}$ denote the size of the state that includes the county ($\mu \leq \mathcal{M}$). Let $\mathcal{S}$ denote the home bias of the state, which we found to be 0.61 and is at least as large as the home bias of the county, so that $\alpha \leq \mathcal{S}$. If demand increases by \$1 in a random county that is part of the state, then the county keeps α and exports $1 - \alpha$. The exports will be absorbed by the other counties of the same state with probability $(\mu/(1 - \mu))((\mathcal{M} - \mu)/\mu)$. The first term $\mu/(1 - \mu)$ is the probability a random county anywhere in the whole economy absorbs the exports, while $(\mathcal{M} - \mu)/\mu$ is the relative size of the state area, excluding the original county that received the dollar. As a result, we can write the state-level home bias using the formula

$$\mathcal{S} = \alpha + (1 - \alpha) * \frac{\mu}{1 - \mu} \frac{\mathcal{M} - \mu}{\mu}$$

Based on population measures, we calculate the relative size of the state as $\mathcal{M} = 0.10$. This implies a county-level home bias equal to $\alpha = 0.58$.

Table 2 Statistics over liquid assets and MPC

Statistic	SCF	
	(1998-2007)	Model
Households with $a/y < 1\%$	0.29	0.30
Liquid assets/income	1.20	1.20
Liquid assets Gini	0.93	0.75
Median MPC		0.11
Average MPC		0.37

Notes: The table presents summary statistics regarding wealth and the MPC. All statistics are reported at an annual frequency.

1.2 Steady-state results

In Table 8, we compare the model to data with respect to the liquid asset distribution. We also report the median and the average MPC. Our model is calibrated to capture the average liquid asset-to-income ratio as well as the fraction of households with asset-to-income ratios less than 1%. The wealth Gini is 0.75 in our model, which is lower than the empirical value of 0.93 but somewhat higher than in typical Aiyagari models. Table 8 also reports the median and the average MPC. We transform the quarterly into an annual MPC using the formula $1 - (1 - \text{quarterly MPC})^4$. The annual average MPC in our model is 0.37, which corresponds to a quarterly MPC of 0.11.

There has been ample recent evidence on the magnitude of consumption responses to unexpected income transfers. Most studies find annual estimates of MPC between 0.2

and 0.6 (?). Our MPC estimate is within this range. Sahm et al. ? analyse survey responses and find that roughly one-third of the 2008 economic stimulus rebate income was spent and that the spending was concentrated in the few months after the receipt.⁴ Also analysing the tax rebates of the 2008 economic stimulus, Parker et al. ? find that during the first three months following payment, households spend between 12 and 30 cents of every dollar received on nondurable goods. Once durable spending is taken into account, the authors find a MPC of between 50 and 90 cents per dollar. Our quarterly MPC estimate is at the lower bound of their estimates regarding non-durable spending. Jappelli and Pistaferri ? use a survey that asks how much people would consume or save were they unexpectedly to receive a transfer equal to their monthly income. They find substantial heterogeneity, with the average MPC being around 48%. Similar to the evidence provided by Jappelli and Pistaferri ?, in our model, households with the lowest net worth exhibit the highest MPCs.

1.3 The government spending shock and transition dynamics

We analyse the effect of a government spending shock on consumer spending. Figure 4 shows average county-level Recovery Act spending between 2009 and 2012.⁵ We approximate the process using an AR(2) for the government spending shock: $G_t = (1 - \rho_1 - \rho_2) G_{ss} + \rho_1 G_{t-1} + \rho_2 G_{t-2}$. Parameters ρ_1, ρ_2 are chosen to match the county-level spending in the data. We pick

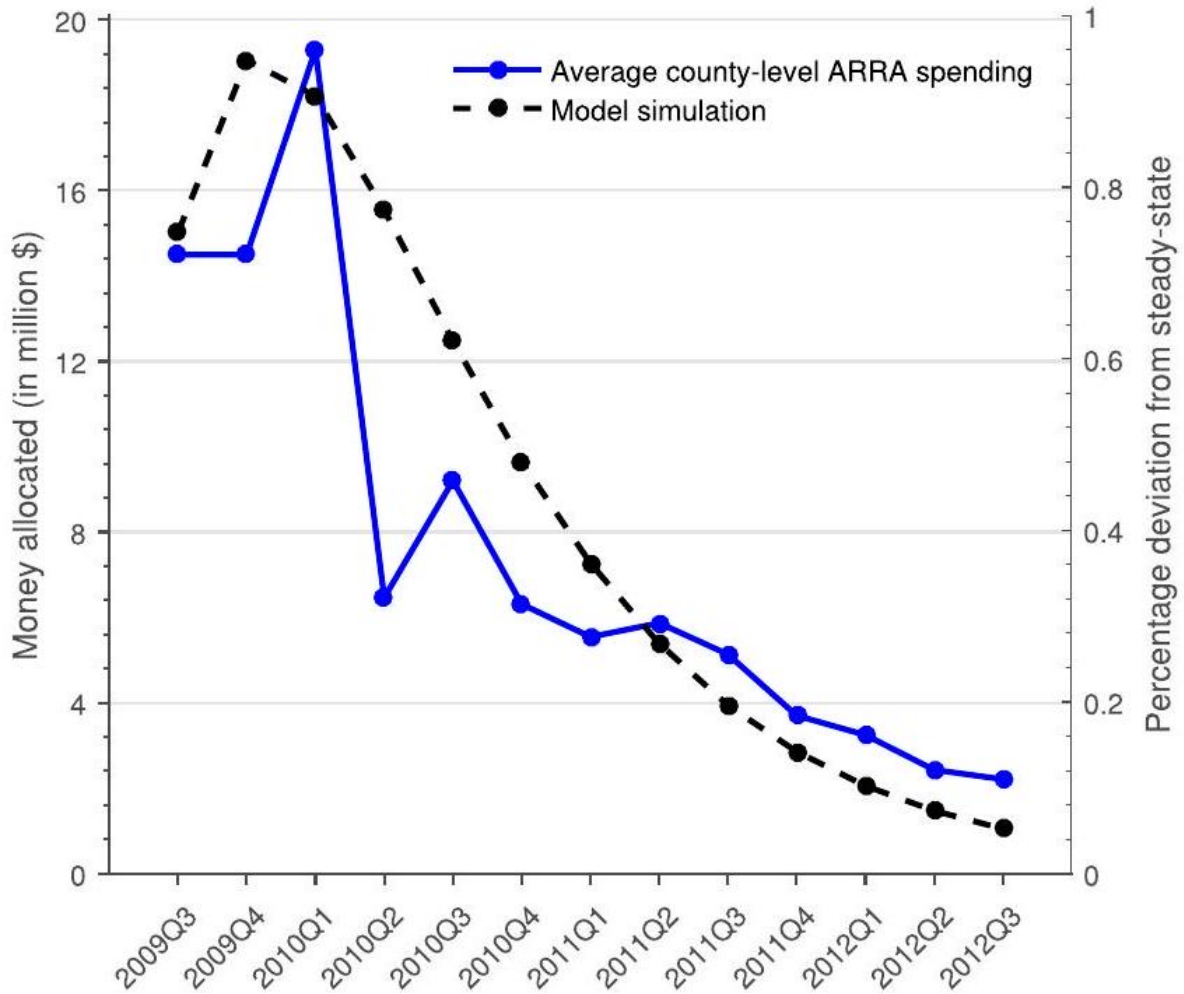
Notes: Average county-level Recovery Act spending (in millions of dollars), by quarter (left axis). Model simulation of government spending shock for region 1 (right axis).

the impact shock in region 1 so that the peak of the simulated path is 1% higher than the steady state. We calibrate the shock for region 2 to be 36% lower than the shock in region 1 since percapita spending at the 25 th percentile of the distribution of ARRA funds was around 36% lower than at the 75th percentile. The shock is assumed to be a one-time unexpected innovation, and households can perfectly foresee the future evolution of prices and quantities.

We plot the impulse response functions for macroeconomic aggregates in Figure 5. All quantities are expressed in per capita terms. The increase in government spending G_1 increases the demand for final good Y_1 . As a result, local inflation π_1 increases (upper right panel). To accommodate the extra demand, intermediate good firms in region 1 demand more labour, which increases the local real wage w_1 (middle left panel). The percentage increase in labour income turns out to be higher than the percentage increase in total income so that dividends decrease (middle right panel).

⁴Their estimate of one-third is likely biased downward as the propensity to consume is bounded at one when they translate the "mostly spend" responses to an MPC. Sahm et al. ? document that the way the stimulus is delivered (reduction in tax withholdings versus one-time payments) matters for how much households spend.

⁵We do not have data for projects in 2009, so we use numbers from Uhlig (2010). However, this does not affect our calculations since these projects show up in cumulative spending of the following years, provided they did not finish before the first quarter of 2010.



Recovery Act Spending: Data versus Model

Per-capita government spending in region 2 is less than per-capita government spending in region 1. However, due to trade linkages, a fraction of the stimulus spreads to region 2 in the form of higher demand for intermediate inputs. As a result, inflation in both regions responds almost equally.⁶ Wages w_2 also increase in region 2, both due to the local fiscal stimulus and the increased demand for local inputs coming from region 1.

Higher inflation in region 1 relative to region 2 implies an initial small appreciation of the real exchange rate $Q_{i'i} = P'_i/P_i$. The appreciation induces an expenditure switching effect. The final good firm in region 1 substitutes local with cheaper foreign intermediate inputs. This tends to make the comovement of economic aggregates between the regions even higher. Moreover, federal taxes adjust to keep the budget balanced. However,

⁶Both the aggregate and the relative response of inflation are small due to the relatively high degree of price stickiness. We discuss the empirical validity of our calibration in Section 5.4.

due to higher inflation, which decreases some of the government’s debt service cost, the need to adjust taxes is relatively small.

Notes: Impulse response functions for a temporary government spending shock. All units are in per-capita terms and are expressed as percentage deviations from their steady state. For the inflation rate, we report the deviation from the steady state in levels.

This redistribution of resources from the private to the public sector hurts net savers, namely, wealthy, low-MPC households. In contrast, the small adjustment in taxes affects a broader group of consumers including low-income, high-MPC households.

Both regions increase consumer spending as a response to fiscal stimulus (lower left panel). Region 1 consumes more than region 2 not only on impact but throughout the transition. This happens because region 1 saves some of its higher income during the fiscal stimulus, while region 2 deaccumulates asset holdings (lower right panel).

Table 3 Consumption multipliers: data versus model

Horizon	$t = 8$	$t = 16$	$t = 32$
<i>Data</i>			
Local	-	0.29	-
<i>Model</i>			
Local	0.15	0.20	0.26
Aggregate	0.41	0.41	0.47

Notes: The empirical target of 0.29 is the consumption multiplier for spending on non-durable goods and services (Table 5).

Notes: Consumption change and multipliers due to wage, dividends, inflation, and taxes. Each case sets a variable to its equilibrium path and all others to their steady-state value. The table reports four-year consumption multipliers. Equilibrium paths are shown only for region 1.

We compute local and aggregate consumption multipliers using the model-generated impulse responses. The local consumption multiplier is computed from the model-generated regional data using the same specification as in our empirical analysis (see equation (3.2)). The aggregate multiplier is computed as $\sum_i \mu_i \Delta C_{i,t} / \sum_i \mu_i \Delta G_{i,t}$, where $\Delta C_{i,t}$ and $\Delta G_{i,t}$ denote the cumulative change of consumption and government spending, respectively, in region i and in year t , relative to the steady state.

Table 9 presents our main two findings. First, the model generates a positive local multiplier reasonably close to the empirical target. Second, we find an aggregate fiscal multiplier equal to 0.41.

Table 10 decomposes the change in consumer spending due to wages, dividends, inflation, and taxes. In particular, we set a variable equal to its equilibrium path and assume that the other variables remain constant at their steady-state values. This generates the marginal effect of a variable on total consumer spending. The left panel plots the consumer spending path for every case as well as when all effects are considered (Benchmark). The table reports the fouryear consumption multiplier in each case as well as the total effect. The total effect does not necessarily equal the sum of the individual

effects due to interaction effects. We consider only region 1 , but the effects for region 2 are qualitatively similar.

As mentioned, wages increase along the transition path as a response to higher demand for labour. Higher labour income affects mainly low wealth, labour-income-dependent households who have high MPCs and increase substantially their spending. If only wages had changed, the local multiplier would be 0.28 while the aggregate would be 0.48 . On the other hand, the decrease in dividends hurts mainly low-MPC households who are less responsive. If only dividends had changed, the local multiplier would be -0.05 while the aggregate would be -0.06 . The combined effect of the two yields an aggregate multiplier of 0.42 .

If only inflation had changed, the local multiplier would be -0.02 and the aggregate multiplier -0.08 . The contribution of inflation to consumer spending is quantitatively small relative to the contribution of wages.⁷

If only tax rates had changed, the local multiplier would be zero since taxes occur at the federal level. The aggregate multiplier is equal to 0.05 . In the first years of the fiscal stimulus, tax rates actually decrease slightly. The reason is that higher inflation decreases the debt service cost of the government.

Our heterogeneous agents, incomplete markets model generates positive consumption responses both at the local and the aggregate levels. In Section 6.1, we show that an incomplete markets, representative agent, new Keynesian (RA-NK) model also generates positive local and aggregate multipliers, albeit much smaller than our Benchmark. In contrast, in models with complete markets, the local consumption multiplier is negative (see, for example, ?; ?; ?). In addition, we show in Section 5.6 that the positive aggregate consumption multiplier is related to the weak response of monetary policy to fiscal stimulus.

1.4 Empirical evidence on labour income and inflation

The model generates a strong positive local effect of government spending on labour income and a moderate effect on inflation. It is informative to evaluate empirically the effect of the fiscal stimulus on these two variables. We collect information on county-level labour income from the Quarterly Census of Employment and Wages (QCEW). We collect information for inflation from the Bureau of Labour Statistics (BLS). We have information on price indices for the period 2008-14 for 382 Metropolitan Statistical Areas.

The Recovery Act had a positive effect on county-level labour income (left panel; Figure 6 and Table 11). In contrast, there is no effect of government spending on inflation (right panel; Figure 6 and Table 11). Our model is consistent with these patterns. The local effect of government spending on inflation is nearly zero as both regions increase inflation by almost the same amount (Figure 5). This result arises due to our relatively

⁷We are not the first to stress the relative stronger effect of wages and the relative weaker effect of inflation on consumer spending. See, for example, the analysis in Kaplan et al. ?.

high degree of price stickiness. Moreover, the local multiplier associated with labour income is positive, consistent with the empirical evidence.

1.5 The role of trade linkages

Figure 7 shows the local and the aggregate consumption multiplier as we vary the degree of home bias α and keep all other parameters same as in the benchmark model. Consumption multipliers are reported at a four-year horizon. In our calibration, we set home bias equal to 0.58 using

Notes: We vary parameter α which captures the degree of home bias for region 1. The benchmark calibration is $\alpha = 0.58$ (vertical line). We show the local and the aggregate consumption multiplier for a four-year horizon.

Table 4 Responses of labour income and inflation to government spending (2008-12)

Variable data source	Labour income QCEW		Inflation BLS	
	OLS	IV	OLS	IV
Government spending	1.33*** (0.34)	0.81* (0.41)	-0.000 (0.001)	-0.0002 (0.002)
Partial F stat.	-	78.6	-	166.5
County controls/state fixed effects	Yes	Yes	Yes	Yes
# of counties	2,916	2,916	1,116	1,116

Notes: The first two columns show estimates of a regression of percentage change in labour income on cumulative government spending at the county level during the period 2008-12. The last two columns show estimates of a regression of percentage point change in inflation on log-cumulative government spending at the county level during the period 2008-12. We show results for our OLS and IV specification and the standard errors in parentheses. One, two, and three stars denote significance at the 10%, 5%, and 1% levels, respectively.

Information on shipments of goods from the CFS (vertical line). This corresponds to a local consumption multiplier equal to 0.20 and an aggregate consumption multiplier equal to 0.41.

The local consumption multiplier is zero when $\alpha = \mu_1$. When home bias increases, trade flows between the two regions decline and region 1 keeps more of the local fiscal stimulus. As a result, consumption in region 1 increases more relative to consumption in region 2. Thus, our model is consistent with the empirical evidence presented in Section 3.4: lower home bias (stronger trade linkages) decreases the local consumption multiplier. The aggregate consumption multiplier is largely unaffected by the strength of trade linkages. Since the two regions are (percapita) symmetric, the final division of government spending does not matter for the aggregate consumption response.⁸

⁸When α is closer to one, the increase in government spending increases substantially demand for goods and inflation in region 1 and generates expected deflation going forward. Deflation increases the

Notes: We vary parameter α which captures the degree of home bias for region 1. The benchmark calibration is $\alpha = 0.58$ (vertical line). We show the local and the aggregate consumption multiplier for a four-year horizon.

1.6 The role of monetary policy

We analyse how monetary policy affects the response of consumption to fiscal stimulus. The nominal interest rate is given by a Taylor rule: $R_t = R_{ss} + \zeta \hat{\pi}_t$. For our benchmark parametrization, we have $\zeta = 0$, which corresponds to a monetary policy that is unresponsive to inflationary pressures, a case resembling the ZLB episode. Figure 8 shows the local and aggregate four-year consumption multiplier when we vary the Taylor rule coefficient ζ and keep all other parameters as in the benchmark parametrization.

The aggregate multiplier decreases gradually as we increase the responsiveness of the monetary authority to aggregate inflation. Consumer spending drops for two reasons. First, the increase in the nominal rate increases the real interest rate, depressing consumer spending. This is true for both regions as there is a currency union. Second, the increase in the nominal rate increases the government debt service cost. As a result, to balance the budget, the government increases taxes. The combined effect of higher real interest rates and higher taxes decreases consumer spending.

In contrast, the local consumption multiplier is largely unaffected by the responsiveness of monetary policy. This exercise confirms the intuition of the literature claiming that first, monetary authority is critical for the value of the aggregate fiscal multiplier (?), and second, monetary policy does not affect the local multiplier so that local estimates are an upper bound for the multiplier during times of conventional monetary policy (see, for example, ?).

1.7 Using the local multiplier to inform the aggregate multiplier

Our model maps the empirical evidence on the local multiplier to an estimate for the aggregate multiplier. In this section, we show that when the model is also informed by the local multiplier,

Notes: The Taylor rule is given by $R_t = R_{ss} + \zeta \hat{\pi}_t$. In Benchmark, we have $\zeta = 0$. We vary parameter ζ and report the four-year local and aggregate consumption multiplier.

it delivers a tighter range of estimates for the aggregate multiplier relative to a model that is only informed by the MPC.

Specifically, for selected model parameters, we change the parameter value to a plausible alternative (one parameter at a time) and report the average MPC, the local multiplier, and the aggregate multiplier. Figure 9 shows the model's multipliers and the MPC for three different values of each of the following parameters-the dispersion of the discount factor, dividend tax, elasticity of substitution, dividend allocation, lump-sum transfer, and duration of the government spending shock.

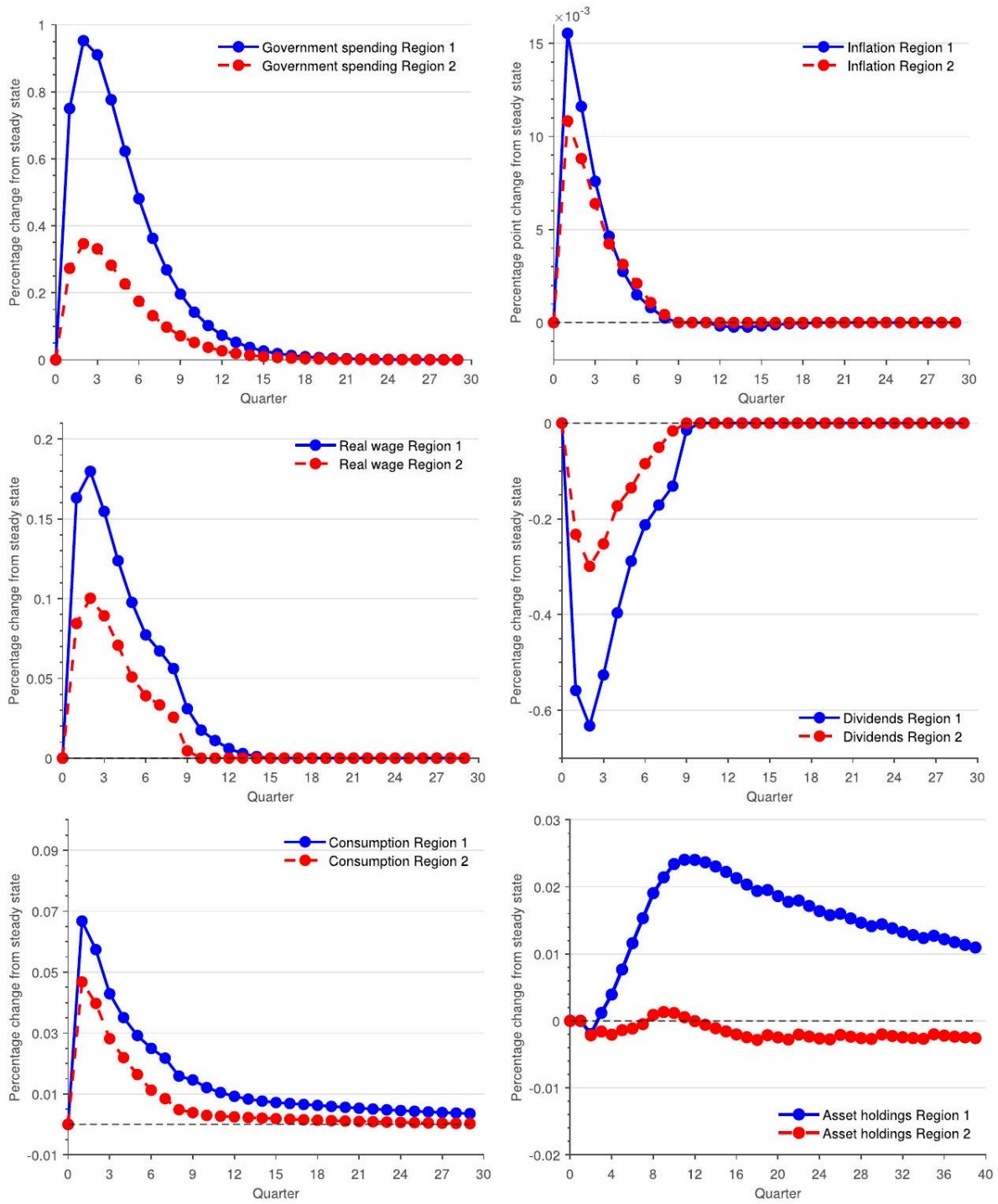
real interest rate, inducing consumers to save, and therefore depresses consumption. This explains why the local consumption multiplier decreases when home bias approaches one.

First, the figure shows that the local and the aggregate multiplier vary in a systematic way. For the most part, larger values of the local multiplier are associated with larger values of the aggregate multiplier. Hence, the local multiplier is a useful predictor of the aggregate multiplier.

Second, and more importantly, the local multiplier is a useful statistic even if we take into account the value of the MPC. To demonstrate this, Figure 10 shows the model's MPC, local, and aggregate multipliers from Figure 9 in one scatterplot. Each dot represents a combination of the MPC and the local multiplier (with the MPC on the x -axis and the local multiplier on the y -axis), with the value of the aggregate multiplier listed next to the dot. The values of the MPC between 0.35 and 0.45 are considered within the range of reasonable empirical estimates. As can be seen from the figure, for this range, the aggregate multiplier varies substantially, from 0.29 to 0.56. This demonstrates that the MPC is not the only factor influencing the aggregate multiplier. To pin down the value of the aggregate multiplier, we can use the local multiplier. Even conditional on MPC, the local and the aggregate multiplier vary in a systematic (positive) way. A value of the local multiplier equal to or above 0.20 -which is in line with our empirical estimates-narrows the range of the aggregate multiplier to 0.41 – 0.56. Thus, the local multiplier is informative about the aggregate multiplier beyond the estimates of the MPC.

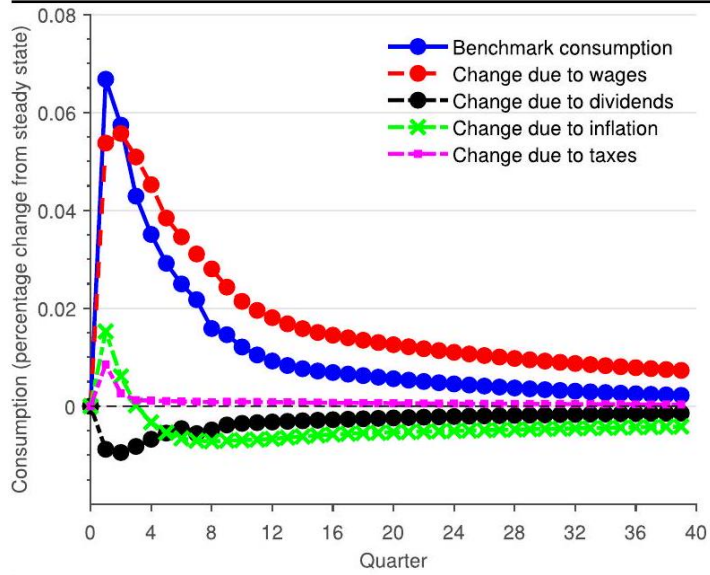
Notes: We vary a parameter away from its benchmark value and explore how the MPCs and the multipliers respond. The vertical dashed line shows the benchmark value for the parameter. For the duration of government spending case, government spending equals the initial benchmark amount for a given number of quarters and then is set to zero. We do not recalibrate the other parameters to hit the benchmark targets.

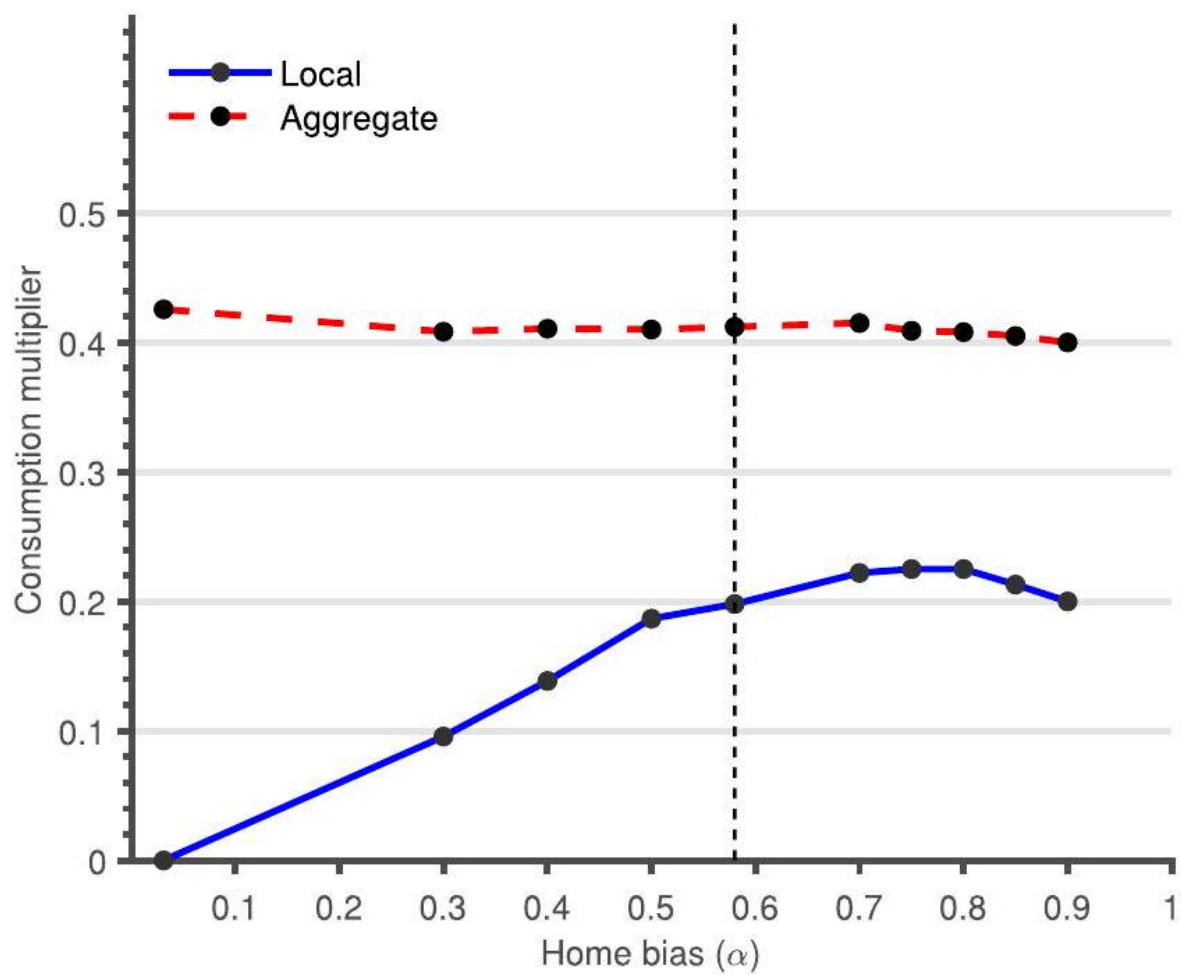
Notes: The figure shows a scatter of the MPC and the local multiplier under different specifications: (a) discount factor dispersion, (b) dividend tax, (c) elasticity of substitution, (d) dividend allocation, (e) lump-sum transfer, and (f) duration of government spending. We denote the high parameter values of each specification with a prime. Next to each specification we report the value of the aggregate multiplier.



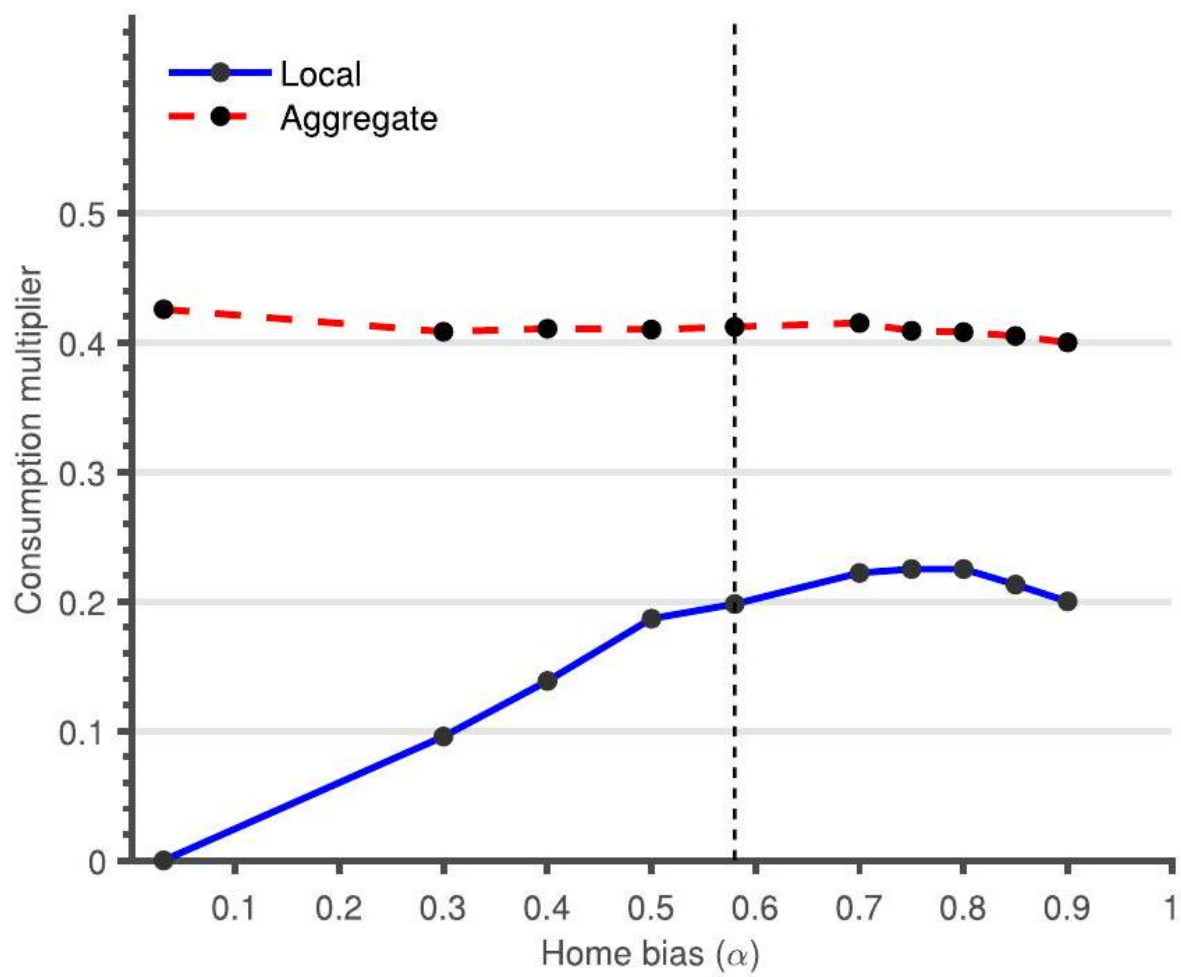
Impulse responses to a government spending shock

TABLE 10
Change in consumption: decomposition

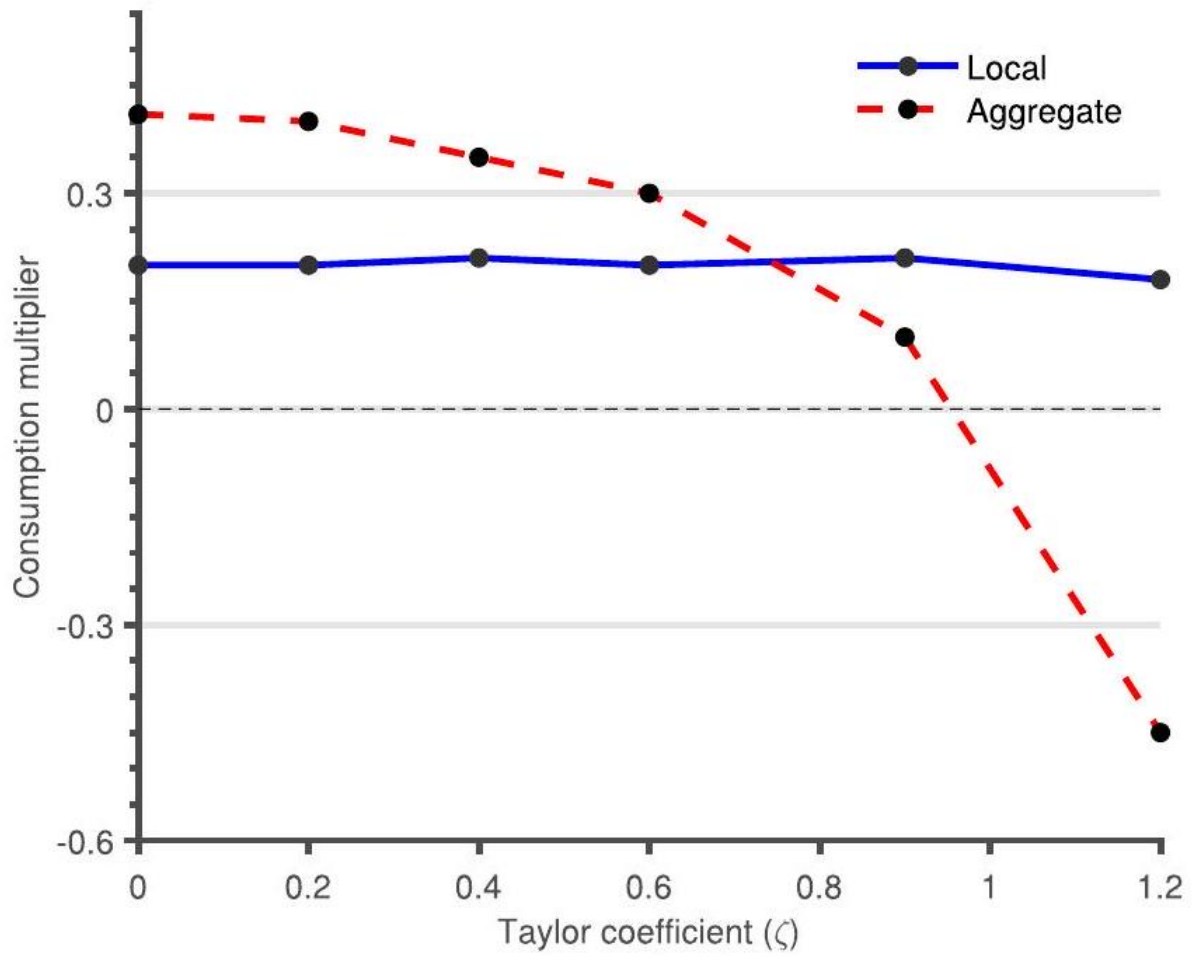




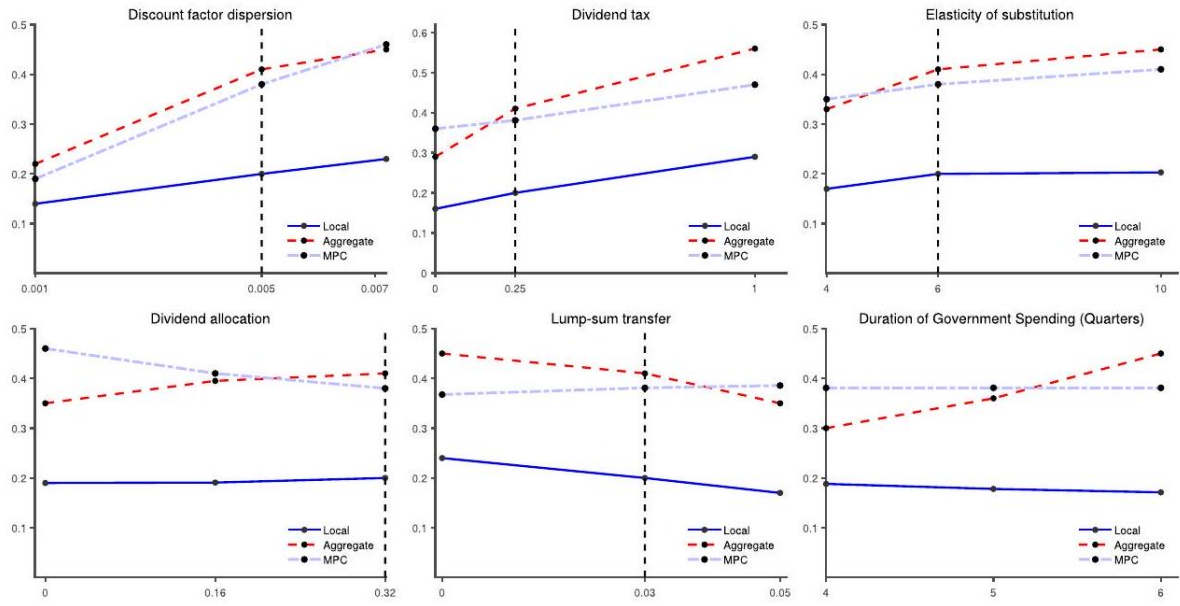
Consumption multipliers and trade linkages (model)



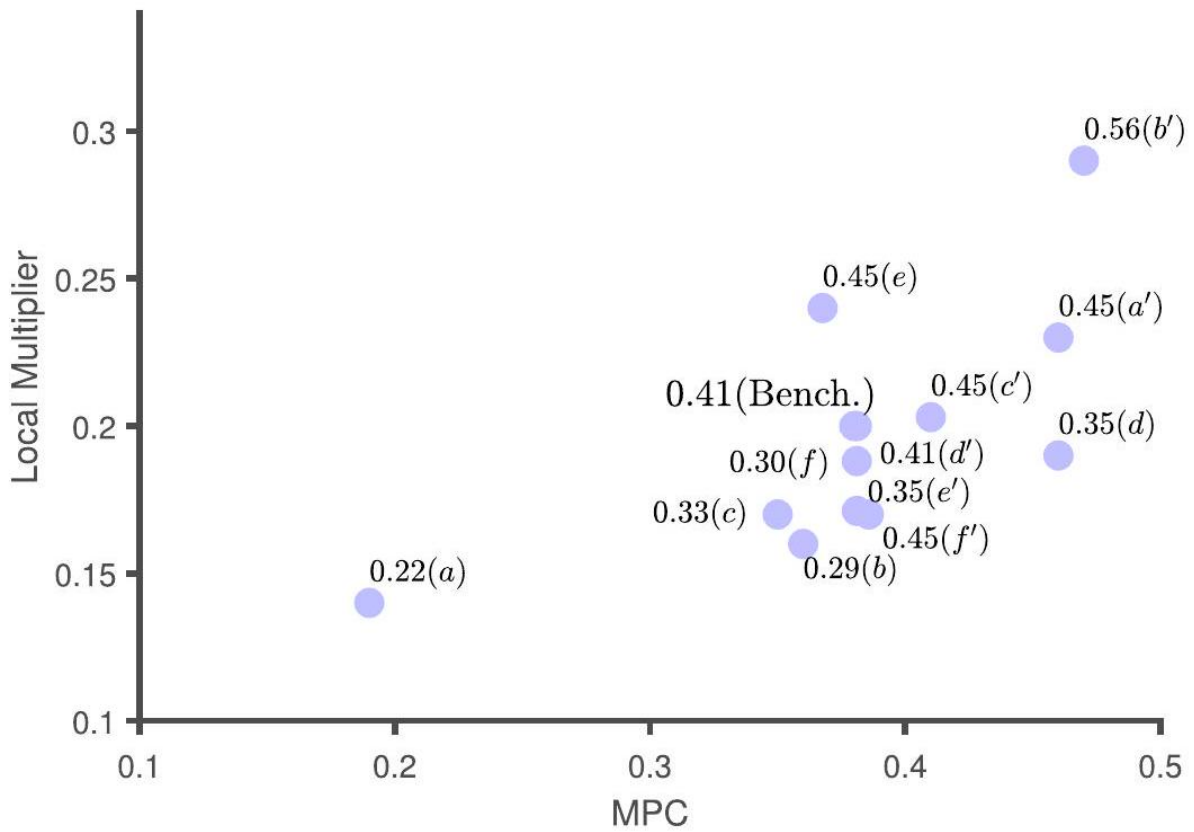
Consumption multipliers and trade linkages (model)



Consumption multipliers and monetary policy



Sensitivity analysis: MPC, local, and aggregate multiplier



MPC, local, and aggregate multiplier under different specifications